MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS)
THIRD SEMESTER EXAMINATION
(2024 ADMISION ONWARDS)
MG3DSCMAT202-ESSENTIALS OF APPLIED MATHEMATICS

Duration: 2 hrs Maximum Marks: 70

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

Part A
Answer any five questions. Each question carries 2 marks

1. Find the real and imaginary part of $z_1 z_2$ where $z_1 = 8-3i$ and $z_2 = 9+2i$. [U][1]
2. Find the value of $\frac{1}{i^6}$ [U][1]
3. Write Bessel's equation [K][2]
4. What is the order of the defferential equation $\frac{d^3y}{dx^3} + (\frac{dy}{dx})^2 = y^3 + \sin x$ [U][2]
5. Find the cartesian coordinates of the point $(2, \frac{2\pi}{3})$ [U][3]
6. Write the polar equation for the ellipse with eccentricity e and semimajor axis a. [K][3]
7. Find the velocity and accelaration of the particle whose motion in space is given by the position vector $r(t) = 2\cos t i + 2\sin t j + 5\cos^2 t k$. [A][4]
8. Define the directional derivative of a differentiable function on the plane. [K][4]

Part B
Answer any five questions. Each question carries 6 marks

9. Write the exponential form of the complex number $z = -1 - i$ [A][1]
10. Find the derivative of $f(z) = 3z^2 - 2z + 4$ [U][1]
11. Determine the singular points of the following functions [A][1]

    (a) $f(z) = \dfrac{2z+1}{z(z^2+1)}$

(b) $f(z) = \dfrac{z^3 + i}{z^2 - 3z + 2}$

12. Solve the differential equation $xy' = (1 - 2x^2 \tan y)$ [A][2]

13. Find the cartesian equivalent of the polar equation $r = 1 - \ cos\theta$. [K][3]

14. Find the standard form equation of the ellipse with center origin, focus $(\sqrt{5}, 0)$ and the corresponding directrix $\frac{9}{\sqrt{5}}$. [U][3]

15. Find unit tangent vector, principal normal and curvature of the curve $r(t) = a\cos ti + a\sin tj + btk$, $a, b \geq 0$ and $a^2 + b^2 \neq = 0$ [A][4]

16. Find the equation of tangent to the ellipse $\frac{x^2}{4} + y^2 = 2$ at the point $(-2, 1)$. [A][4]

Part C
Answer any three questions. Each question carries 10 marks

17. Show that $u(x, y)$ is harmonic in some domain and find a harmonic conjugate $v(x, y)$ when

    (a) $u(x, y) = 2x(1 - y)$

    (b) $u(x, y) = \sinh x \sin y$

    [A][1]

18. Solve

    (a) $xy' - 3y = x^4$

    (b) $e^y dx + (xe^y + 2y)dy = 0$

    [E][2]

19. (a) Solve $xy'' - y' = 3x^2$

    (b) Solve $(y - x^3)dx + (x + y^3)dy = 0$

    [E][2]

20. (a) Find the foci, vertices and center of the ellipse $\frac{x^2}{2} + \frac{y^2}{16} = 1$.Also sketch the ellipse [U][3]

    (b) Show that the equation $x^2 - 4y^2 + 2x + 8y - 7 = 0$ represents a hyperbola. Finds its vertices, foci, center and asymptotes.

    [U][3]

21. Find the graph of the osculating circle of the parabola $y = x^2$ at the origin. [A][4]

22. (a) Find the length of the indicated portion of the curve $r(t) = 4\cos ti + 4\sin tj + 3tk$, $0 \leq t \leq \frac{\pi}{2}$.

    (b) Find the directions in which $f(x, y) = x^2 + xy + y^2$ increases most rapidly and decreases most rapidly at the point $(-1, 1)$.

    [A][4]